Privacy-Enhanced Synthetic Data Generation for Autonomous Vehicle Training

R26-CS-010

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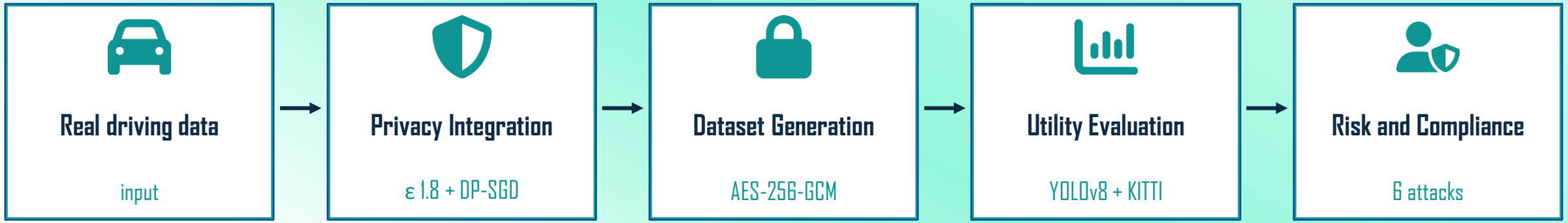
PP2 COMMENTS AND ACTIONS TAKEN

Component	Panel Comment	Action Taken
Privacy Integration	Attack reduction with DP is not significant	Stress test on an overfitted baseline: attack AUC 0.628 to 0.501 (5 seeds, $p \leq 0.001$)
Dataset Generation	Is the original file removed by crypto-shredding?	No: the original is retained, not crypto-shredded. It is securely converted to an encrypted form, with RBAC-managed keys.
Utility Evaluation	How was data poisoning detection validated?	Two planted defects were flagged; the clean dataset passed
Risk and Compliance	Metadata leakage findings are 0	Expected for simulator data; a planted positive control is next

THE PROBLEM AND OVERALL SOLUTION

THE PROBLEM: Synthetic driving data looks private, but real information leaks in through simulator calibration, model training and stored files, and no one outside the team can verify it.

OUR SOLUTION



DESIGN EXCELLENCE OVER EXISTING SOLUTIONS

- Privacy during generation, not after
- Every claim tested by attacks
- Signed, independently verified results
- Utility measured on real data



IT22309556 | S. A. S. D. PRIYADARSHI

Individual Component: **Privacy Integration**

Formal differential privacy at both points where real data enters the pipeline.

NOVELTY: Privacy enforced during data generation rather than post-processing.

PROBLEM

Enforcing and validating differential privacy in the AV data pipeline

THE PROBLEM

Simulator tuned with real driving data

Trained models can leak members

THE GAP

No formal guarantee in AV platforms

Budgets rarely tested by attacks

MY SOLUTION

Laplace noise on calibration (ε 1.8)

DP-SGD training (ε 1, 3, 8)

Attack it to prove it holds

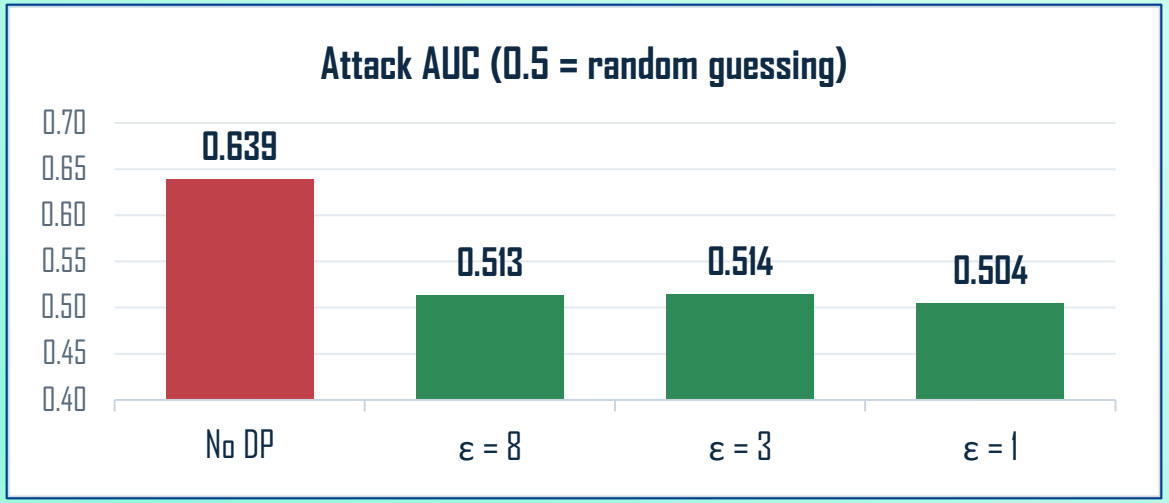
HOW IT WORKS



PRIVACY INTEGRATION

Final Results

PROTECTION	ε	MODEL AUC	ATTACK AUC
None	-	0.83	0.639
Weak	8.0	0.67	0.513
Medium	3.0	0.66	0.514
Strong	1.0	0.62	0.504



COMPLETED

Calibration budget fixed: ε 1.8 total

Stress test: AUC 0.628 to 0.501 ($p \leq 0.001$)

Signed privacy certificate PDF

Attack falls to random guessing (0.5) at a cost of 0.16 to 0.21 model AUC.



IT22541284 | YUBITHA S.

Individual Component: **Secured Dataset Generation**

Privacy-aware multi-sensor datasets from CARLA, encrypted and fingerprinted inside the pipeline.

NOVELTY: Privacy-aware synthetic multi-sensor datasets generated directly from the simulator.

PROBLEM

Protecting synthetic sensor files inside the pipeline

THE PROBLEM

Generated files sit readable on disk

Changes can go unnoticed

THE GAP

Protection added after generation

No integrity proof at handover

MY SOLUTION

AES-256-GCM inside the pipeline

SHA-256 manifest for handover

6-test security audit

HOW IT WORKS

CARLA capture — 5 towns

→

SHA-256 manifest — fingerprint

→

AES-256-GCM — encrypt

→

Plaintext removed — after encrypt

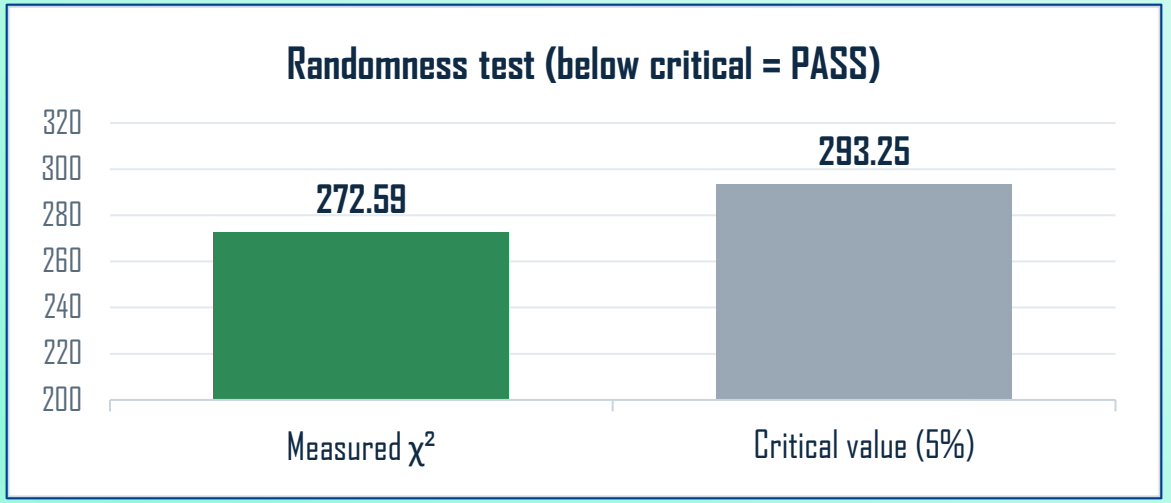
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Secure handover — verified handoff

SECURED DATASET GENERATION

Final Results

TEST	EVIDENCE	RESULT
Round trip	Identical bytes	PASS
Nonces	30 of 30 unique	PASS
Randomness	χ^2 272.59 < 293.25	PASS
Bit difference	50.0% of bits differ	PASS
Tamper	MAC check failed	PASS



COMPLETED

Five towns captured and encrypted
SHA-256 manifests verified (40,000 files)
6 of 6 security audit tests passed

0 of 6,093 captured files readable without the key.



IT22110220 | M.K.S.R. APEKSHA

Individual Component: **Utility Evaluation**

Measures whether privacy-protected synthetic data still trains a useful object detector.

NOVELTY: Framework generating privacy – enhanced synthetic datasets evaluating AV perception privacy – utility trade – off.

PROBLEM

Is privacy-protected synthetic data still useful?

THE PROBLEM

Privacy may leave data too weak

Labels from another party trusted

THE GAP

Utility rarely tested on real data

No poisoning check on labels

MY SOLUTION

Train YOLOv8n on synthetic data

Test on real KITTI images

Poisoning gate before training

HOW IT WORKS

Decrypt + verify

40,000 files

→

Poisoning gate

15,084 boxes

→

YOLOv8n training

synthetic

→

KITTI test

real images

→

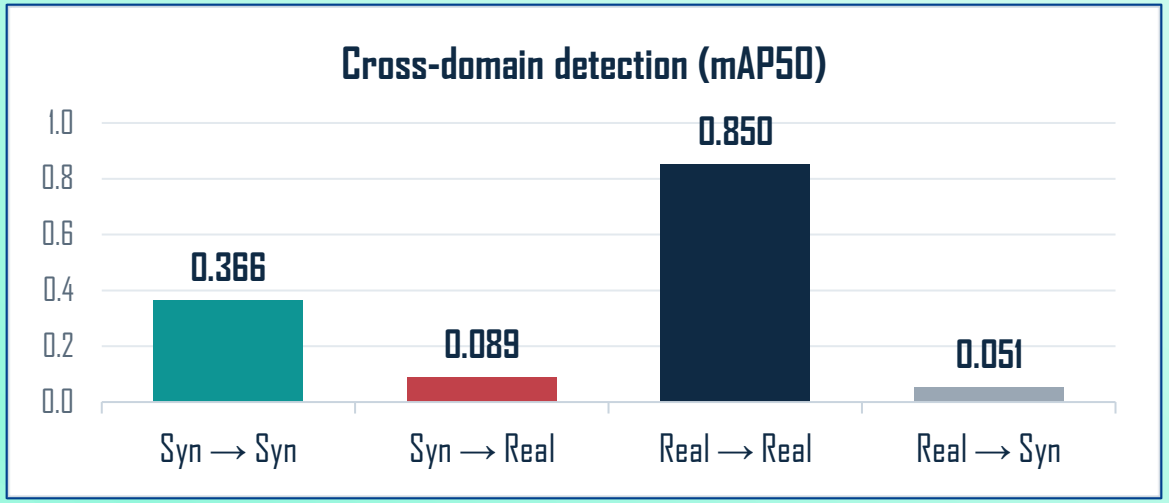
FID

115.99

UTILITY EVALUATION

Final Results

TRAIN → TEST	mAP50	CAR	PEDESTRIAN
Syn → Syn	0.366	0.608	0.179
Syn → Real	0.089	0.261	0.005
Real → Real	0.850	0.952	0.777
Real → Syn	0.051	0.127	0.008



COMPLETED

40,000 files verified; poisoning gate PASS
4 cross-domain experiments on KITTI
FID 115.99: moderate domain gap

Synthetic to real keeps only 10.46% of mAP50.



IT22066916 | R.D.P.T. WIJESINGHE

Individual Component: **Risk and Compliance**

Independent attack-based verification of every privacy and security claim.

NOVELTY: An attack-based privacy evaluation method for synthetic AV data pipelines.

PROBLEM

Verifying privacy claims independently

THE PROBLEM

Teams grade their own claims

Reports can be edited after signing

THE GAP

Few claims tested independently

Scans lack positive controls

MY SOLUTION

Six independent attacks

Re-run attacks and compare

Signed compliance certificate

HOW IT WORKS

Signed reports

from each component

→

6 attacks

own code

→

Compare

within about 2%

→

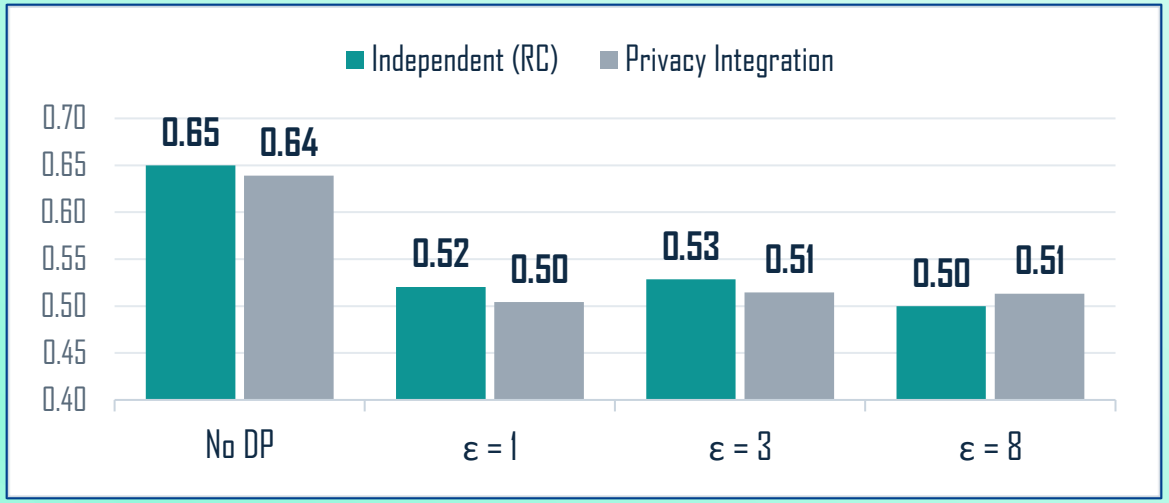
Certificate

SHA-256 + HMAC

RISK AND COMPLIANCE

Final Results

ATTACK	RESULT	STATUS
1 Certificates	4 of 4 verified	PASS
2 Budget audit	Within 0.0001	PASS
3 Reconstruction	Ratio 0.94	PASS
4 Independent MIA	AUC 0.50 to 0.53	PASS
5 Metadata	0 findings	PASS
6 Re-identification	0% matched	PASS



COMPLETED

6 of 6 attacks passed
Replication within about 2%
Signed compliance certificate PDF

Independent results agree with Privacy Integration within about 2%.

RESULTS, LIMITATIONS AND FUTURE WORK

0.504

Attack AUC at ε 1 (no DP 0.639)

6 / 6

Security tests passed

10.46%

mAP50 kept on real KITTI

6 / 6

Independent attacks passed

LIMITATIONS

Weak transfer to real images

Plaintext deleted, not crypto-shredded

FUTURE IMPROVEMENTS

Domain adaptation with real data

In-memory encryption and positive controls



Thank You!